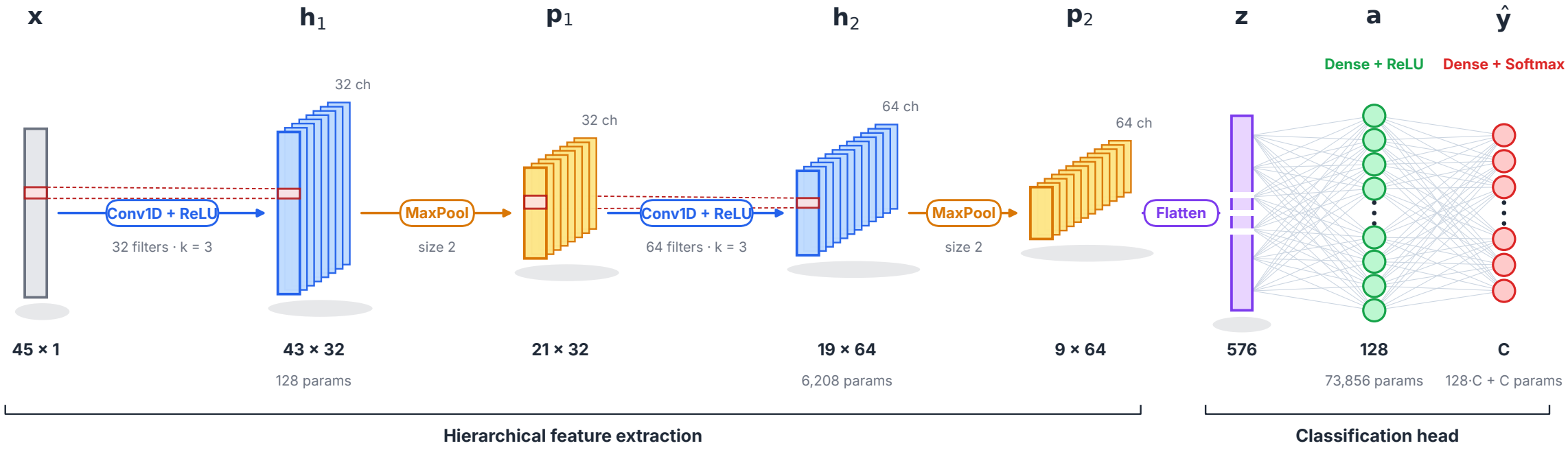
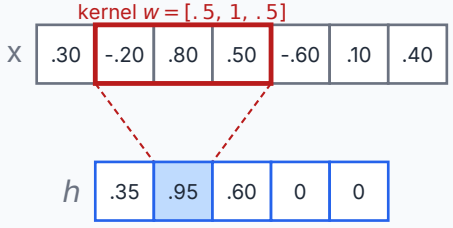
(a) CNN architecture

block heights not to scale



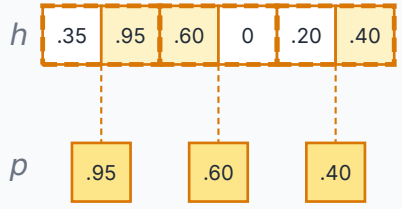
(b) 1-D convolution (k = 3)



$$h_i = \mathrm{ReLU}\left(\sum_{k=1}^{3} w_k x_{i+k-1} + b\right)$$

slides along the 45 features, sharing weights

(c) Max pooling (size 2)



$$p_i = \max(h_{2i},\, h_{2i+1})$$

halves the length, keeps the strongest response

(d) Prediction head

$$\hat{\mathbf{y}} = \mathrm{softmax}(W_2\,\mathrm{ReLU}(W_1\mathbf{z} + \mathbf{b}_1) + \mathbf{b}_2)$$

C = 2	C = 6	C = 19
Benign / Attack	attack families	attack types

Optimiser: Adam · Loss: categorical cross-entropy

10 epochs · batch size 32

one network, three label granularities